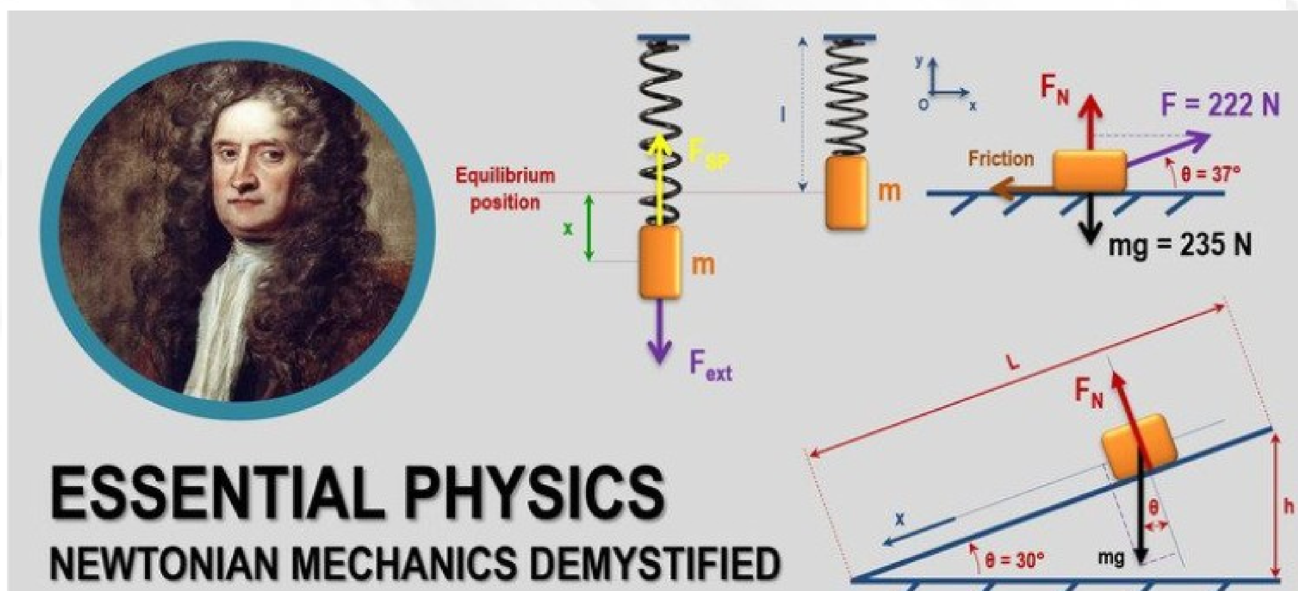
Welcome
Physics 2048
Fall 25

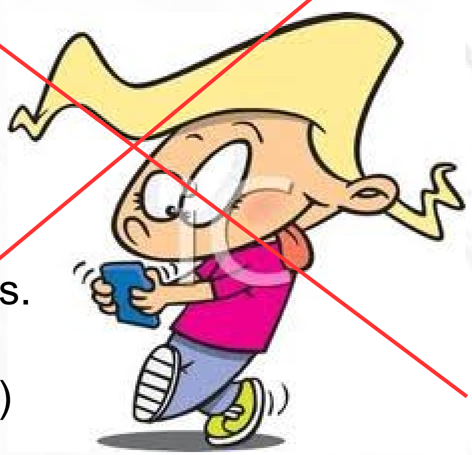


Physics with Calculus I
PHYS 2048   Fall 25

→ Slides, tutorials for HW, solutions for HW can be found in my dropbox (see links in canvas – Lets visit canvas)

→ Syllabus, assignments, link to resources can be found
In canvas.

→ I record the lectures and upload them to my Utube Channel. I still highly recommend to attend in person.



Relative weight will be assigned as follow:

Homework 18 % (Tutorials in dropbox. Prep for test. Don’t cheat)
Tests 60% (test1, test2, final are in person. On paper. 20% each)
pop-up quizzes 12% (pop quiz at the end of each class. Open notes. Discussion with peers encouraged. Based on previous lecture)
take home test 10% (test3. Online. Window of 2 hours. Open notes)
Attendance 5% → **5 points bonus!**

Contact: vlankar@fiu.edu

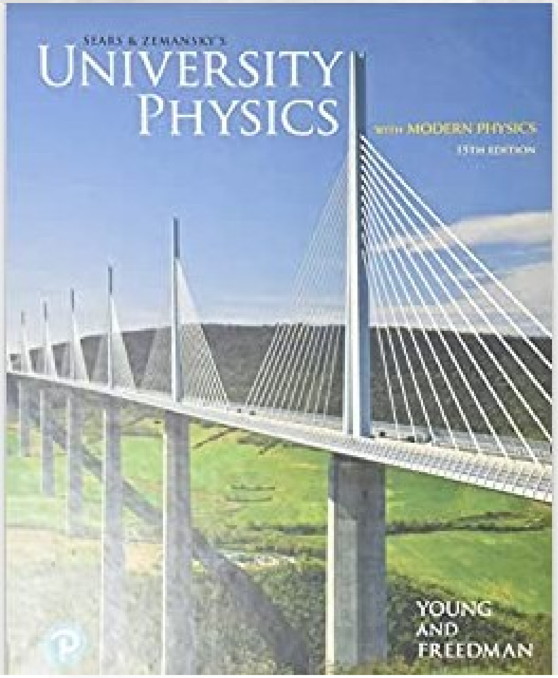
TikTok

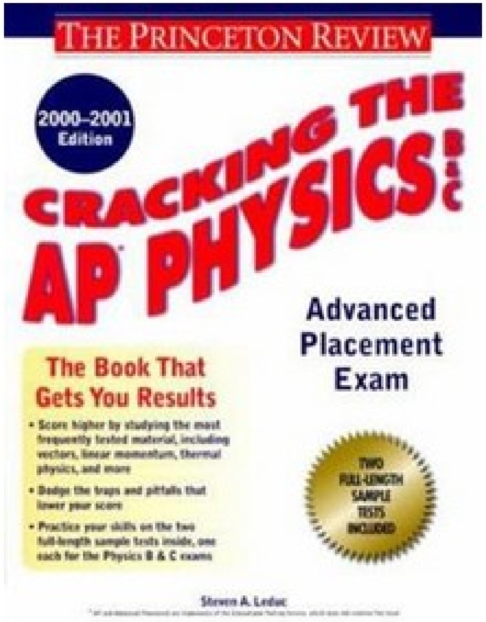
Lectures: Tu/Th 10:35AM-12:15PM @Lib265
Office hours: Tu/Th 12:30PM after class and M/W 11:15AM sharp @ MS150

Bring a TI83/84 or at least a TI 30 in class

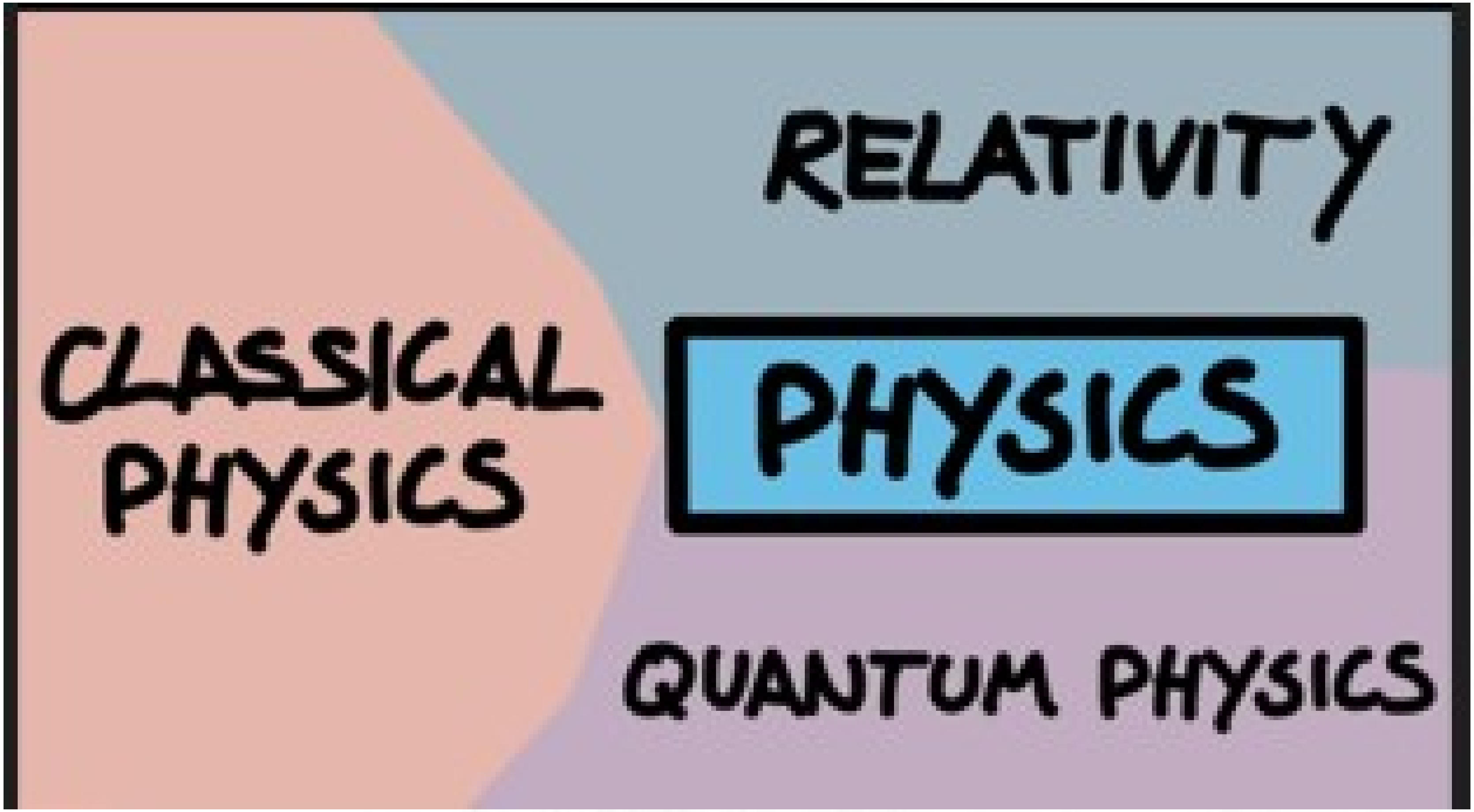
TOPICS for lectures

LECTURES	CHAPTERS
Unit 1: Math review Proportions, scientific notation, conversions	
Unit 2: Kinematics 1D, Free-fall kinematics equations, graphical analysis, free-fall	Chapter 2 2.1 → 2.6
Unit 3: Kinematics 2D - Vectors, components of vectors, Projectile motion, circular motion	Chapter 1 1.7, 1.8, 1.9 Chapter 3 3.1 → 3.5
Unit 4: Newton's laws Newton's laws, forces, weight and apparent weight	Chapter 4 4.1 → 4.6
Unit 5 : Application of Newton's law adding vectors, equilibrium, friction, inclined plane, circular motion	Chapter 4 4.1 → 4.6
Unit 6: Work-energy scalar product, kinetic energy, work done on a spring, work-energy theorem	Chapter 5 5.1 → 5.4
Unit 7: Conservation of energy gravitational potential energy, work-energy theorem extended, conservative and non-conservative forces, PE in a spring, power	Chapter 6 6.1 → 6.4 Chapter 7 7.1 → 7.5
Unit 8: Linear momentum. Center of mass collision. Conservation of momentum	Chapter 8 8.1 → 8.6
Unit 9: rotation Angular kinematics. Moment of inertia	Chapter 9 9.1 → 9.6
Unit 10: torques. Rolling.	Chapter 10 10.1 → 10.4
Unit 11: angular momentum Conservation. Kepler's laws/.	Chapter 10 10.5, 10.6
Unit 12: equilibrium, statics	Chapter 11 11.1 → 11.3
Uni13:: gravitation Inverse square law.	Chapter 13 13.1 → 13.5
Unit 14: Fluids See utube playlist Pressure, fluid dynamics, Bernoulli effect	Chapter 12 12.1 → 12.5
Unit 15: Waves – deformation - SHM Simple harmonic motion. Elastic deformation. Young modulus. Shear deformation. Wave properties. Sound See utube playlist	Chapter 11 11.4 Chapter 14 14.1 → 14.5 Chapter 15 15.1 → 15.3 Chapter 16 16.1 → 16.3





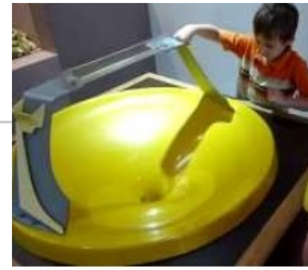
Older edition 2000 - $5

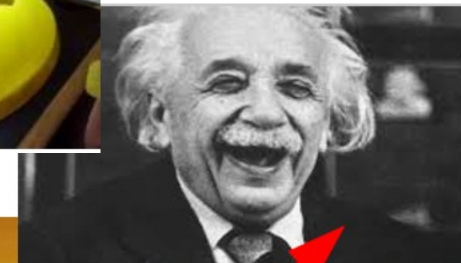


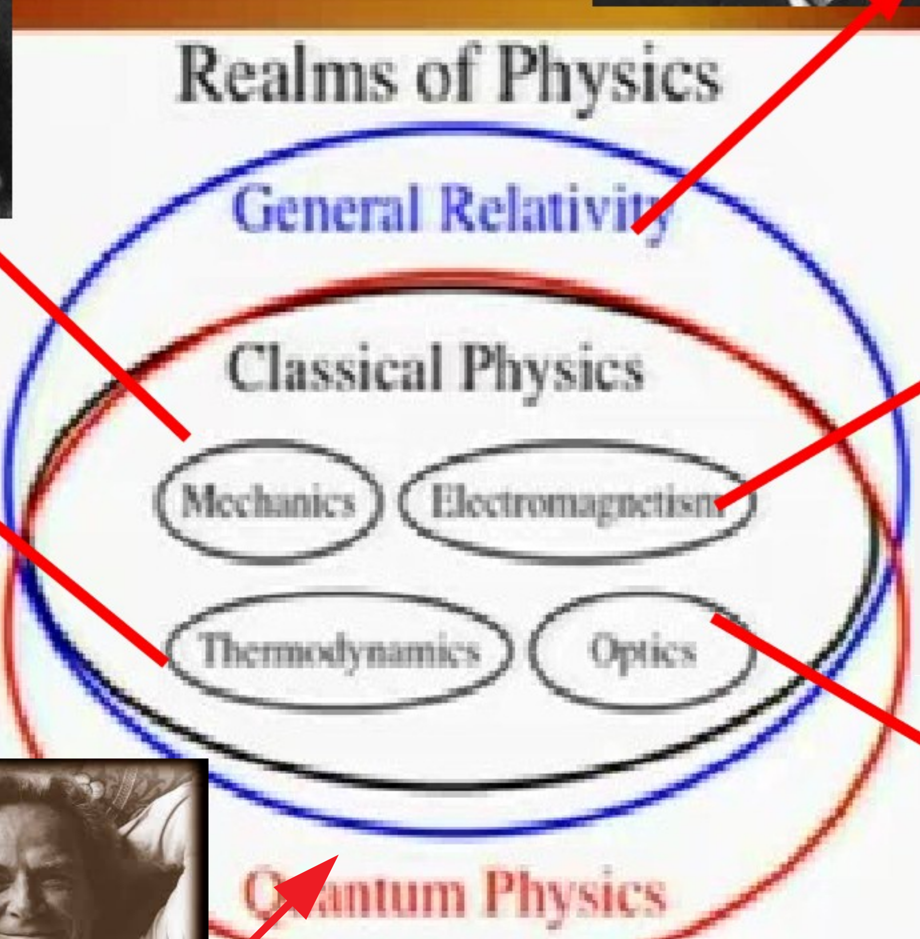
RELATIVITY
CLASSICAL PHYSICS
PHYSICS
QUANTUM PHYSICS

https://www.youtube.com/watch?v=MTY1Kje0yLg

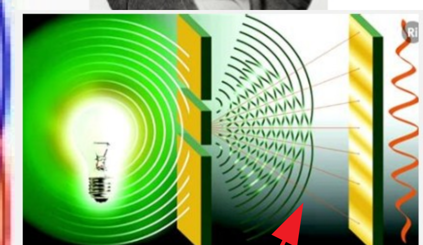
Demo: free-fall
EM /

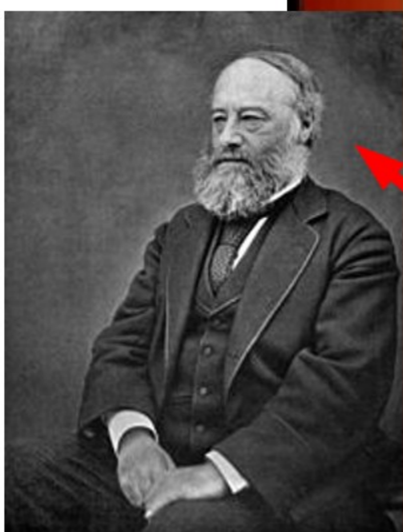


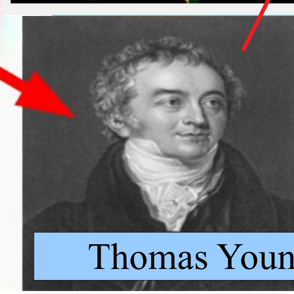




Michael Faraday







Thomas Young

James Prescott
Joule

Richard Feynman

https://youtu.be/Iuv6hY6zsd0?t=281

quiz https://www.youtube.com/watch?v=0NbBjNiw4tk Michio Kaku -watch. Quiz for Friday

Physics experiments involve the measurement of a variety of physical quantities.

These measurements should be accurate and reproducible.

The first step in ensuring accuracy and reproducible is defining the units in which the measurements are made.

Units and derived units

SI Base Units

Base Quantity	Name	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Important:
Speed is in m/s
and m/s is noted m s^{-1}

Acceleration is in m/s/s
And is also noted m s^{-2}

Distance (displacement) is in m

SI Derived Units

Derived Quantity	Name	Symbol	Equivalent SI units
Frequency	hertz	Hz	s^{-1}
Force	newton	N	$m \cdot kg \cdot s^{-2}$
Pressure	pascal	Pa	N/m^2
Energy	joule	J	N·m
Power	watt	W	J/s
Electric charge	coulomb	C	s·A
Electric potential	volt	V	W/A
Electric resistance	ohm	Ω	V/A
Celsius temperature	degree Celsius	°C	K*

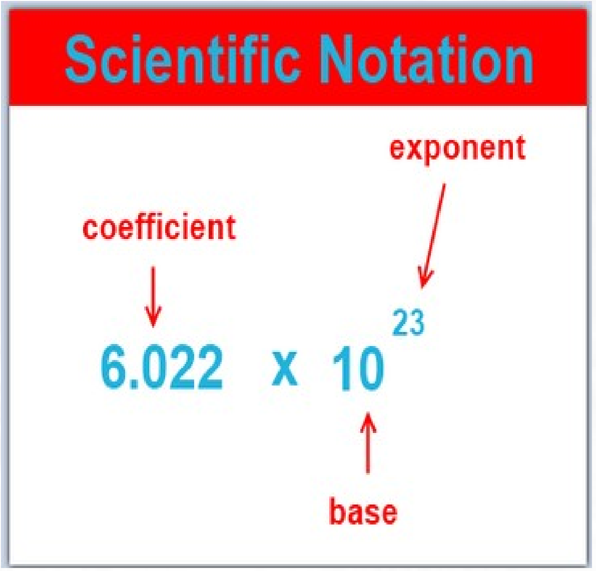
Important:
Units can give away equations.
Force is kg m/s/s →
Find the equation for force

find the equation for power

find the equation for pressure

Find he equation for energy

Scientific notation → assignment_sci_not.pdf



tera (T) = 10^{12}
$5\ TJ = 5 \times 10^{12}$ joules

Following is a list of prefixes and their meanings that are often used in conjunction with SI units:

Multiple	Prefix	Symbol
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n

kilo (k) = 10^{3}
$1\,km = 10^{3}\,m$
$1\,kg = 10^{3}\,g$

mega (M) = 10^{6}
$25\,MB = 25 \times 10^{6}\,B$
B is for bytes

centi (c) = 10^{-2}
$5\,cm = 5 \times 10^{-2}\,cm$
$5\,cg = 5 \times 10^{-2}\,cg$

milli (m) = 10^{-3}
$5\,ml = 5 \times 10^{-3}\,l$
$5\,mm = 5 \times 10^{-3}\,m$
$5\,mg = 5 \times 10^{-3}\,g$

nano (n) = 10^{-9}
$7\,ns = 7 \times 10^{-9}$ sec
$7\,nm = 7 \times 10^{-9}\,m$

micro (μ) = 10^{-6}
$3\,\mu m = 3 \times 10^{-6}\,m$
$3\,\mu g = 3 \times 10^{-6}\,g$

Examples: In the news 8/23/22 295 euros per MWh Unit for energy

Protons in the LHC get about 6.5 tera-electronvolts or TeV.

To print out and to keep.

1km = 1,000m (10^3 meters)
1m = 100cm (10^2 centimeters)
1cm = 10mm (10 millimeters)
1km = 100,000 cm (10^5 cm)
1mm = 1000 μm (10^3 micrometers)
1m=1000 mm (10^3 mm)
1m = 1,000,000 μm (10^6 μm)
note that: if 1m = 10^6 μm then 1μm = 10^{-6} m
1mm= 10^{-3} m 1cm = 10^{-2} m

1 mile = 1,609m
1 mile = 1.6 km
10 N= 2.2 pounds

1min = 60s
1hour = 60min = 3600s
1 day = 24 h = 24x3600 s =86,400s

1kg = 1000 g
1g=1,000mg

Following is a list of prefixes and their meanings that are often used in conjunction with SI units:

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10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n

Remember: 10K run = 6.2 miles about 1 kg is about 2.2 pounds

5K run = 3.1 miles 1m/s is about 2.2 mph (3.6 km/h)

1 mile is about 1.6km (or 1.5) so 100 miles is about 100 + 50 = 150 miles (actually 160)

Metric vs. British Units

1 meter (m) = 39.37 inches

1 kilometer (km) = 1000 m ≈ 0.62 mile

1 centimeter (cm) = 0.01 m ≈ 0.39 inch

453.6 grams (g) = 1 pound

1 kilogram (kg) = 1000 g ≈ 2.2 pounds

1 A.U. 150,000,000 km → how much in miles?

→ Scientific notation?

The Universe by Orders of Magnitude

Size or Distance	(m)	Mass	(kg)	Time Interval	(s)
Proton	10^{-15}	Electron	10^{-30}	Time for light to cross nucleus	10^{-23}
Atom	10^{-10}	Proton	10^{-27}	Period of visible light radiation	10^{-15}
Virus	10^{-7}	Amino acid	10^{-25}	Period of microwaves	10^{-10}
Giant amoeba	10^{-4}	Hemoglobin	10^{-22}	Half-life of muon	10^{-6}
Walnut	10^{-2}	Flu virus	10^{-19}	Period of highest audible sound	10^{-4}
Human	10^{0}	Giant amoeba	10^{-8}	Period of human heartbeat	10^{0}
Highest mountain	10^{4}	Raindrop	10^{-6}	Half-life of free neutron	10^{3}
Earth	10^{7}	Ant	10^{-2}	Period of earth's rotation	10^{5}
Sun	10^{9}	Human	10^{2}	Period of earth's revolution around sun	10^{7}
Distance from earth to sun	10^{11}	Saturn V rocket	10^{6}	Lifetime of human	10^{9}
Solar system	10^{13}	Pyramid	10^{10}	Half-life of plutonium-239	10^{12}
Distance to nearest star	10^{16}	Earth	10^{24}	Lifetime of mountain range	10^{15}
Milky Way galaxy	10^{21}	Sun	10^{30}	Age of earth	10^{17}
Visible universe	10^{26}	Milky Way galaxy	10^{41}	Age of universe	10^{18}
		Universe	10^{52}		

How far is a light-year?

$$1 \text{ light-year} = (\text{speed of light}) \times (1 \text{ year})$$

In scientific notation this is about ? In km ?
1 mile = 1.6 km
Convert to miles and trillion of miles

Speed of light is 3 10^8 m/s = 300,000 km/s

How far is a light-year?

$$1 \text{ light-year} = (\text{speed of light}) \times (1 \text{ year})$$

$$= \left(300{,}000 \frac{\text{km}}{\cancel{\text{s}}}\right) \times \left(\frac{365\ \cancel{\text{days}}}{1\ \cancel{\text{yr}}} \times \frac{24\ \cancel{\text{hr}}}{1\ \cancel{\text{day}}} \times \frac{60\ \cancel{\text{min}}}{1\ \cancel{\text{hr}}} \times \frac{60\ \cancel{\text{s}}}{1\ \cancel{\text{min}}}\right)$$

$$= 9{,}460{,}000{,}000{,}000 \text{ km}$$

In scientific notation this is about ? In km ?
1 mile = 1.6 km
Convert to miles and trillion of miles

Speed of light is 3 10^8 m/s = 300,000 km/s

Video
Speed of light
(see Astro)

Is our galaxy emptier of an atom of hydrogen (1 proton, 1 electron)?

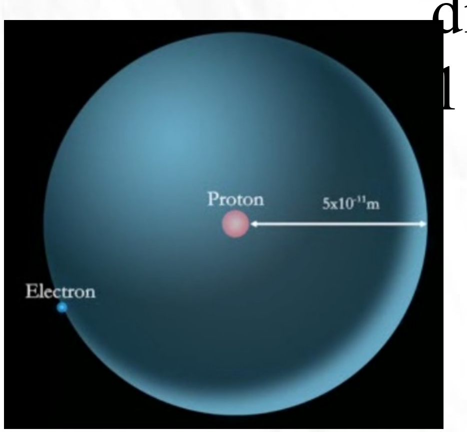
Compare ratio (distance electron/size proton = 510^4) to ratio (distance proxima/ size Sun)

GIVEN the Sun is 100 times larger than the Earth in size

radius of Earth = 6,000 km

distance proxima = 4.2 light year

1 light year = 9.42 trillion km



$\approx 6 \times 10^7 \gg 5 \times 10^4$

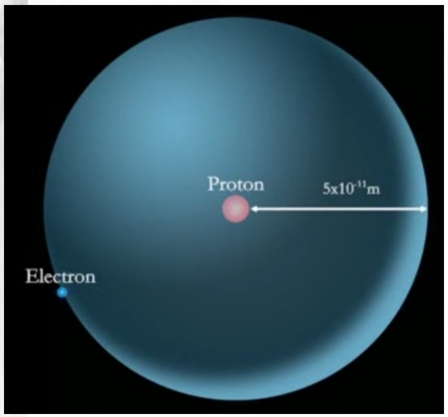
Is our solar system emptier of an atom of hydrogen ?

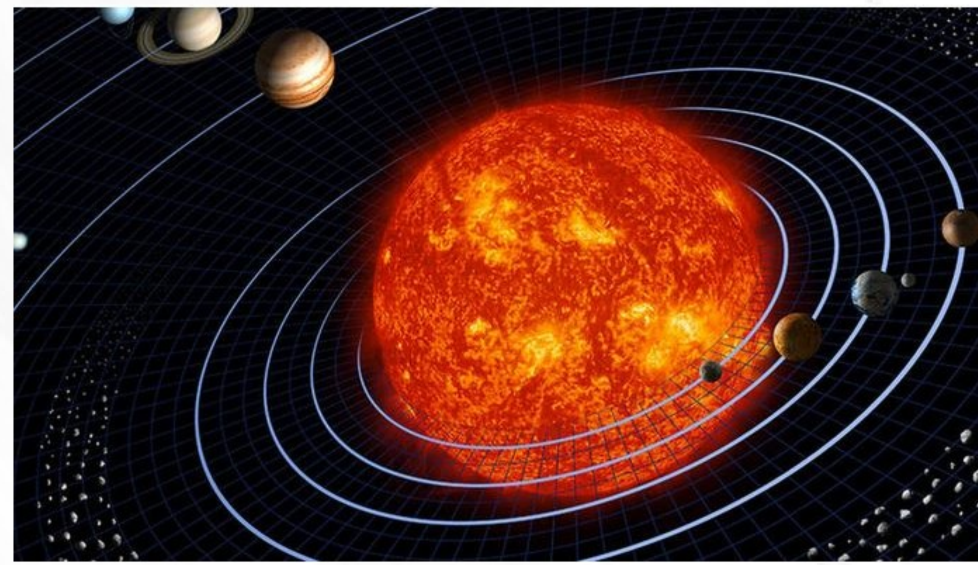
Compare ratio (distance electron/size proton = 510^4) to ratio (distance to Earth/ size Sun)

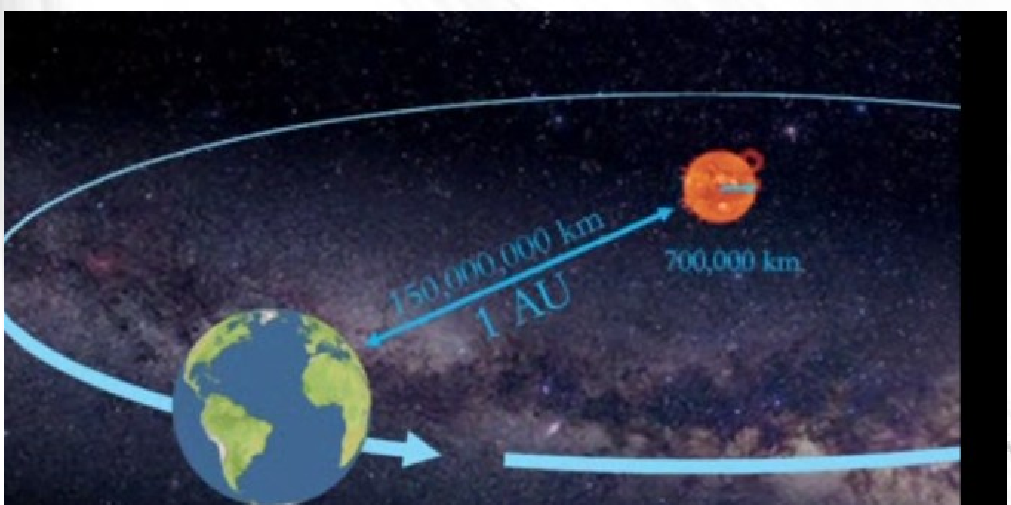
GIVEN the Sun is 100 times larger than the Earth in size

radius of Earth = 6,000 km

distance Earth-Sun = 150,000,000 km







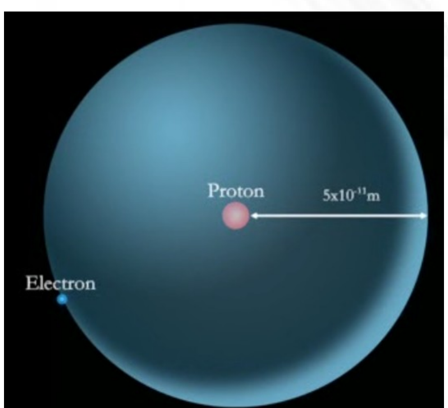
$\approx 200 << 5 \times 10^4$

Is our universe emptier of an atom of hydrogen ?

Compare ratio (distance to electron/size proton = $5 10^4$) to ratio (distance to Andromeda/ size our galaxy)

radius our galaxy = 50,000 light years across

distance Andromeda = 2.5 million light years



How long for light from the Moon to reach us?

Light travels at a finite speed (300,000 km/s).

382,500 kilometers